
EDUCATION**Duke University, Pratt School of Engineering***Bachelor of Science in Mechanical Engineering***Durham, NC***Aug '22- May '26*

- **GPA:** 3.47
- **Engineering Merit Scholarships :** Dorcas Maynor and E. Ray Bucher, Jr. Scholar, Bee-Keng Boey Scholar
- **Relevant coursework:** Aircraft Performance, Compressible Fluid Flow, Heat Transfer.

TECHNICAL SKILLS

Software [SolidWorks(Advanced Assemblies, Fusion 360, ANSYS Fluent, AutoCAD, MATLAB, Python, MS Office (Excel, PowerPoint, Word)]

Engineering [FAA(Private Pilot(Part 61), Part 107), Weight & Balance Analysis, Center of Gravity Estimation, Mass Properties Fundamentals, Bill of Materials (BOM) creation, FEA,CFD(Ansys Fluent)]

Manufacturing [CNC Machining, Injection Moulding, Prototyping, Water Jetting, Shop Floor Support]

ENGINEERING EXPERIENCE**Mechanical Engineering Intern | Trio Labs | Morrisville, North Carolina***May '25 – Aug '25*

- Improved production efficiency by 25%, reducing system downtime and increasing throughput across additive manufacturing(AM) workflows.
- Developed validation protocols for next-gen micro-metal AM systems, accelerating commissioning by 50%.
- Led troubleshooting efforts that reduced recurring mechanical faults by ~20% through root cause analysis and process refinement.

Aeroelasticity Lab Research Assistant |Earl Dowell Lab| Duke University*Apr '25 –May'26*

- Designed and fabricated 5+ mechanical test fixtures, reducing experimental setup time by 30%.
- Executed 20+ wind tunnel tests, validating CFD/XFOIL results within <8% variance.
- Developed structured MATLAB-based analysis scripts, decreasing data processing time by 40%.

Stall Strip Aerodynamics Researcher| Duke University*Aug'25- May'26*

- Designed and tested multiple stall strip geometries, improving stall onset prediction accuracy by ~15% compared to baseline models.
- Modeled components in SolidWorks, reducing assembly misalignment by 20% via tolerance analysis.
- Conducted wind tunnel validation under varied Reynolds number conditions, generating performance datasets across 10+ flight-relevant scenarios.

Lead Flight Test Engineer Electric Glider Performance Improvement Project| **Duke University***Jan '23 - July '23*

- Designed and evaluated structural modifications for a lightweight electric glider, including flap integration, hinge loads, and wing reinforcement using FEA and analytical methods
- Served as Lead Flight Test Engineer, developing test plans, overseeing flight operations, and analysing performance data (stall speed, sink rate) to validate structural and aerodynamic designs.

Hydropower Systems Modelling & Energy Planning Researcher| Duke University*Aug'25- May'26*

- Co-authored research improving hydropower modelling accuracy by ~5% across 15 facilities.
- Analysed multi-basin data, enhancing forecasting resolution for capacity planning.
- Corrected modelling assumptions that impacted long-term grid reliability projections.

Chassis Design and Assembly Engineer | Duke University Motorsports*Aug '23 –May'26*

- Designed 10+ complex structural components in SolidWorks for competitive FSAE vehicle.
- Contributed to vehicle mass optimization efforts, achieving a ~5% reduction in subsystem weight without compromising structural integrity.
- Applied FEA to chassis structures to analyse stress, deformation, and load paths, supporting weight optimization while maintaining structural safety margins..

LEADERSHIP & ACTIVITIES

Duke Cycling: Managed \$20K budget; organized fundraising events and coordinated with vendor.

Languages: French (Fluent), Hindi (Fluent), Kannada (Native)Spanish (Intermediate), Japanese (Beginner)

Interests: Aviation, Ice/Road Speed Skating, Cycling, Cricket, Photography